

✓ Consider a population of 50 units numbered $\{1, 2, \dots, 50\}$.

The only possible samples are the following:

$$\{1\}, \{1, 2\}, \{1, 2, 3\}, \dots, \{1, 2, \dots, 50\}$$

One of these samples is selected at random, and each sample has the same probability of being chosen.

For a unit i , let p_i denote the probability that unit i is included in the selected sample.

For example, if $i = 8$, then p_8 is the probability that the selected sample contains the number 8.

Which of the following statements is correct?

A. $p_i = \frac{i}{50}$

B. $p_i = \frac{50-i}{50}$

C. $p_i = \frac{1}{i}$

D. None of the above



✓ Let X be a random variable which is symmetric about 0. Let F be the cumulative distribution function of X . Which of the following statements is always true?

A. $F(x) + F(-x) = 1$ for all $x \in \mathbb{R}$.

B. $F(x) - F(-x) = 0$ for all $x \in \mathbb{R}$.

C. $F(x) + F(-x) = 1 + P(X = x)$ for all $x \in \mathbb{R}$.

D. $F(x) + F(-x) = 1 - P(X = -x)$ for all $x \in \mathbb{R}$.



✓ Let Y_1, Y_2 be independent random variables taking values -1 and $+1$ with probability $1/2$ each.

Define a sequence $\{X_n\}$ by

$$X_1 = Y_1, \quad X_2 = Y_2, \quad X_n = X_{n-1}X_{n-2} \text{ for } n \geq 3.$$

Find the value of $P(X_4 = 1)$.

- A. $1/4$
- B. $1/2$
- C. $3/4$
- D. 1

✓ Consider the quadratic equation $x^2 + 2Ux + V = 0$, where U and V are chosen independently and at random from the set $\{1, 2, 3\}$, with each value having equal probability.

What is the probability that the given quadratic equation has both roots real?

- A. $2/3$
- B. $1/2$
- C. $7/9$
- D. $1/3$

✓ Suppose X_1, X_2, X_3 and X_4 are independent and identically distributed continuous random variables.

Consider the condition $X_4 > \max(X_1, X_2) > X_3$.

This means:

- X_4 is the largest among X_1, X_2, X_3, X_4
- X_3 is the smallest
- X_1 and X_2 lie in between (in any order)

Find $P(X_4 > X_3 > \max(X_1, X_2))$.

- A. $1/6$
- B. $1/8$
- C. $1/12$
- D. $1/24$

✓ A fair coin is tossed repeatedly.

Let X denote the number of Tails observed before the first Head occurs. Then

$$P(X = k) = P(\underbrace{T, T, \dots, T}_{k \text{ times}}, H)$$

[Let Y denote the number of Tails observed between the occurrence of the first Head and the occurrence of the second Head.

Which of the following statements about the distributions of X and Y are true?

- A. $P(X = k) = 2^{-(k+1)}$ for $k = 0, 1, 2, \dots$, and 0 otherwise.
- B. $P(Y = k) = 2^{-(k+1)}$ for $k = 0, 1, 2, \dots$, and 0 otherwise.
- C. $P(X = k) = 2^{-k}$ for $k = 1, 2, 3, \dots$, and 0 otherwise.
- D. $P(Y = k) = (k+1)2^{-(k+2)}$ for $k = 0, 1, 2, \dots$, and 0 otherwise.

Let $X = (X_1, X_2, X_3)^T$ be a random vector with distribution $N_3(\mu, \Sigma)$, that is, a *three-dimensional multivariate normal* distribution with mean vector μ and covariance matrix Σ .

The mean vector is given by

$$\mu = (1, 1, 1),$$

and the covariance matrix is

$$\Sigma = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & c \\ 1 & c & 2 \end{pmatrix}.$$

Suppose that the random variables X_2 and $-X_1 + X_2 - X_3$ are independent.

Then the value of c is _____.

Assume that $X \sim \text{Binomial}(n, p)$ for some $n \geq 1$ and $0 < p < 1$, and $Y \sim \text{Poisson}(\lambda)$ for some $\lambda > 0$.

Suppose that

$$\mathbb{E}[X] = \mathbb{E}[Y]$$

Then:

- A. $\text{Var}(X) = \text{Var}(Y)$
- B. $\text{Var}(X) < \text{Var}(Y)$
- C. $\text{Var}(X) > \text{Var}(Y)$
- D. $\text{Var}(X)$ may be larger or smaller than $\text{Var}(Y)$, depending on the values of n, p, λ

Let X and Y be independent and identically distributed random variables.

Consider the following statements:

S1: If X has a normal distribution, then the random variable $X + Y$ also has a normal distribution.

S2: If X has a binomial distribution, then the random variable $X + Y$ also has a binomial distribution.

Which of the following is **correct**?

- A. Only statement S1 is correct.
- B. Only statement S2 is correct.
- C. Both statements S1 and S2 are correct.
- D. Neither statement S1 nor S2 is correct.

Let X and Y be two independent random variables, each representing the number of failures preceding the first success in a sequence of Bernoulli trials, with p as the probability of success in a single trial and q as the probability of failure. Then $P(X = Y)$ is:

- A. $\frac{q}{1+q}$
- B. $\frac{p}{1+p}$
- C. $\frac{q}{1+p}$
- D. $\frac{p}{1+q}$

N

Given the probability density function

$$f(x) = \begin{cases} e^{-x}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

find the value of the cumulative distribution function $F(2)$.

- A. $1 + e^{-2}$
- B. e^{-2}
- C. e^{-1}
- D. $1 - e^{-2}$

Let X and Y have joint probability density function given by

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 \leq x \leq 1-y, \quad 0 \leq y \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

If f_Y denotes the marginal probability density function of Y , then $f_Y(1/2) = \underline{\hspace{2cm}}$

Let X and Y be two random variables having the joint probability density function

$$f(x,y) = \begin{cases} 2 & \text{if } 0 < x < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Then the conditional probability $P\left(X \leq \frac{2}{3} \mid Y = \frac{3}{4}\right)$ is equal to

- A. $\frac{5}{9}$
- B. $\frac{2}{3}$
- C. $\frac{7}{9}$
- D. $\frac{8}{9}$

Let $X_1, X_2, \dots, X_n$ be independent exponential random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$.

Define $Y = \min(X_1, X_2, \dots, X_n)$.

The random variable Y has an exponential distribution with parameter λ .

Which of the following statements is/are correct?

A. $P(Y > t) = P(X_1 > t, X_2 > t, \dots, X_n > t)$

B. $P(Y > t) = P(X_1 > t) P(X_2 > t) \cdots P(X_n > t)$

C. $\lambda = \lambda_1 + \lambda_2 + \cdots + \lambda_n$

D. $\lambda = \max\{\lambda_1, \lambda_2, \dots, \lambda_n\}$

N

Let $\mathbf{X} \sim \mathcal{N}_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, i.e., $\mathbf{X} = (X_1, X_2, X_3)^\top$ is a three-dimensional multivariate normal random vector with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.

The mean vector is

$$\boldsymbol{\mu} = (1, 1, 1),$$

and the covariance matrix is

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & c \\ 1 & c & 2 \end{pmatrix}.$$

Find the value of c such that the random variables X_2 and $-X_1 + X_2 - X_3$ are independent